

wk14: EXERCISE -- Pandas Aggregating -- Summarizing and Grouping

- Due Thursday by 3pm
- Points 200
- Submitting a file upload
- File Types pdf and csv
- Available Dec 2 at 6pm - Dec 4 at 3pm

This assignment was locked Dec 4 at 3pm.

As we did in class, use **Pandas** to do the following calculations.

For this exercise, use **YOUR own unique, randomly generated Excel CSV file containing 100 transactions**.

(See the instructions included with the file for creating this file.)

Calculate:

- **Totals for All Transactions**
 - **Total Scoops** for all 100 transactions as a whole
 - **Average scoop size** for all 100 transactions as a whole
- **Conditional Subsetting with Row Values**
 - **Flavor:**
 - **Total number of transactions** that are **Chocolate** row
 - **Total Scoops** for just the **Chocolate** rows
 - **Average Scoops** for just **Chocolate** rows
 - **Total number of transactions** that are **Vanilla** rows
 - **Total Scoops** for just the **Vanilla** rows
 - **Average Scoops** for just **Vanilla** rows
 - **Total number of transactions** that are **Blue Moon** rows
 - **Total Scoops** for just the **Blue Moon** rows
 - **Average Scoops** for just **Blue Moon** rows
 - **Container:**
 - **Total number of transactions** that are **Waffle Cone** rows
 - **Total Scoops** for just the **Waffle Cone** rows
 - **Average Scoops** for just **Waffle Cone** rows
 - **Total number of transactions** that are **Sugar Cone** rows
 - **Total Scoops** for just the **Sugar Cone** rows

- **Average Scoops** for just **Sugar Cone** rows

- **Groupings**
 - **Total number of transactions by flavor**
 - **Total Scoops by flavor**
 - **Average Scoops by flavor**

 - **Total number of transactions by container**
 - **Total Scoops by container**
 - **Average Scoops by container**

Submit both of the following:

- a **PDF** of your work here that contains:
 - your Pandas **code**,
 - your Pandas **console output**,
 - your data spreadsheet **100** transactions that were generated at random
(Copy and paste from your CSV file in Excel into your Word document.)

- your **CSV** file that you used
 - This must be the unique CSV that you generated, consisting of 100 transactions generated at random

Programming Exercise Rubric - 200

Criteria	Ratings	Pts
Code is correct and complete. Code is correct. Code is very similar to the kind of code modeled in class and on the lecture-demo SLIDES. Code is pasted as TEXT, not as a screen snapshot. Code is not the result of ChatGPT or other AI or copied and pasted from another student.		150 pts
Screen Snapshots or Copy of Output Screen Snapshots/Copies of Output are correct. Code is similar to the Python/Pandas code style and techniques used during in-class demonstrations and on the slides. Screen snapshots are clear, correct, and well-formatted.		35 pts
Student name is in the code and output and filename. Student name is included in both CODE and screen snapshots of output. Student NAME is in the PDF file name. Example: LECTURE-DEMO_2-10_JoeWisco_v1z.pdf		15 pts
Total Points: 200		